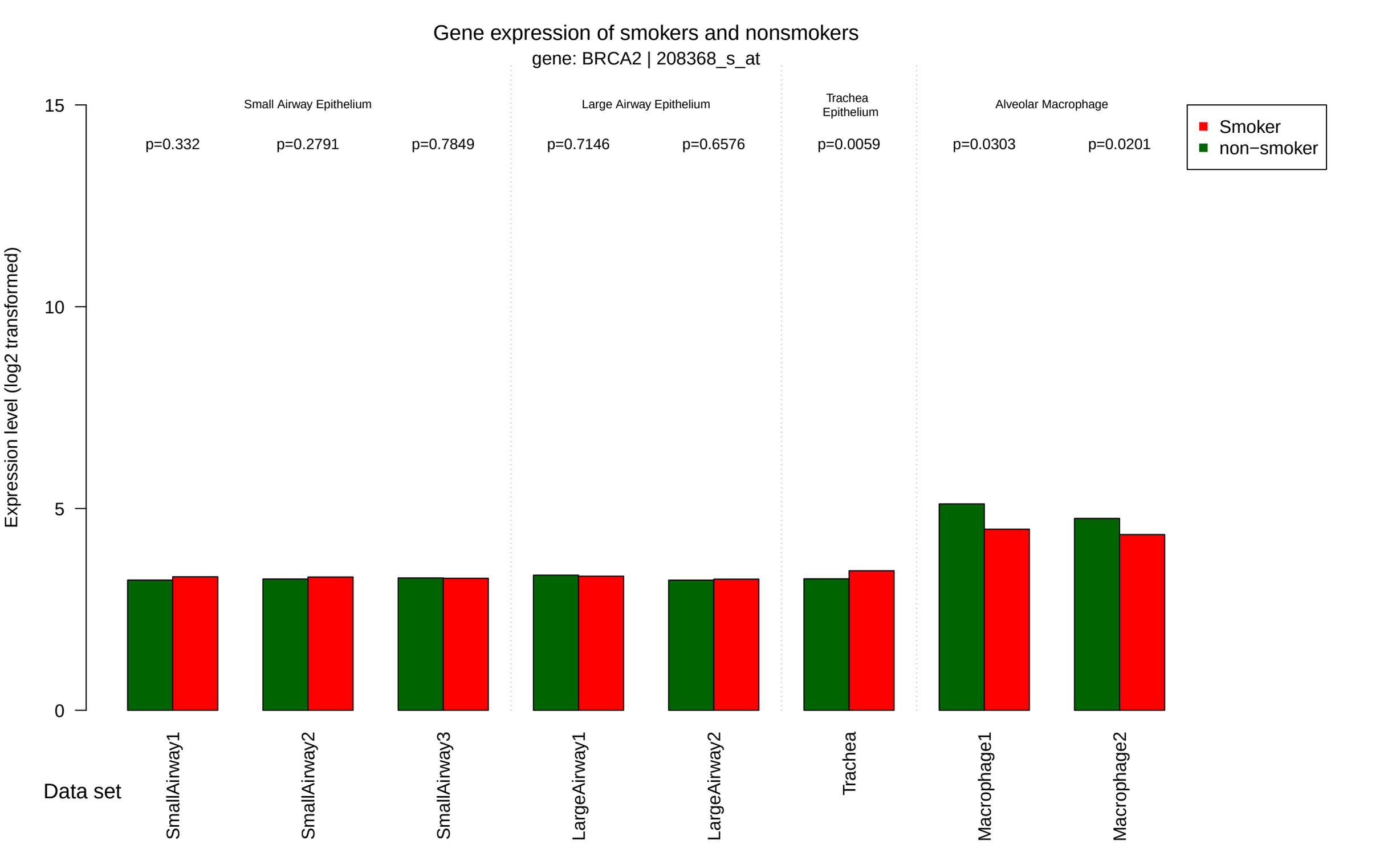
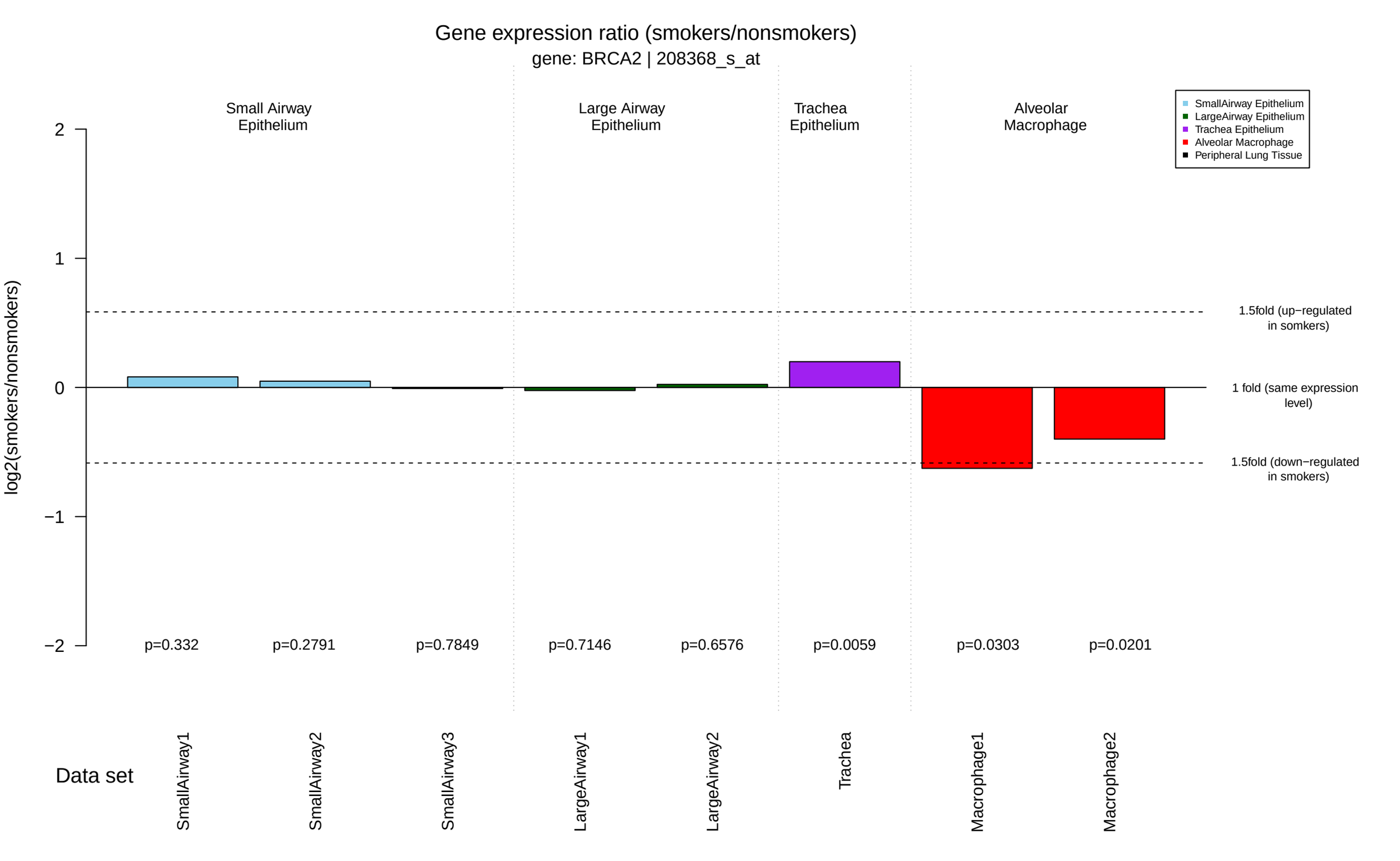
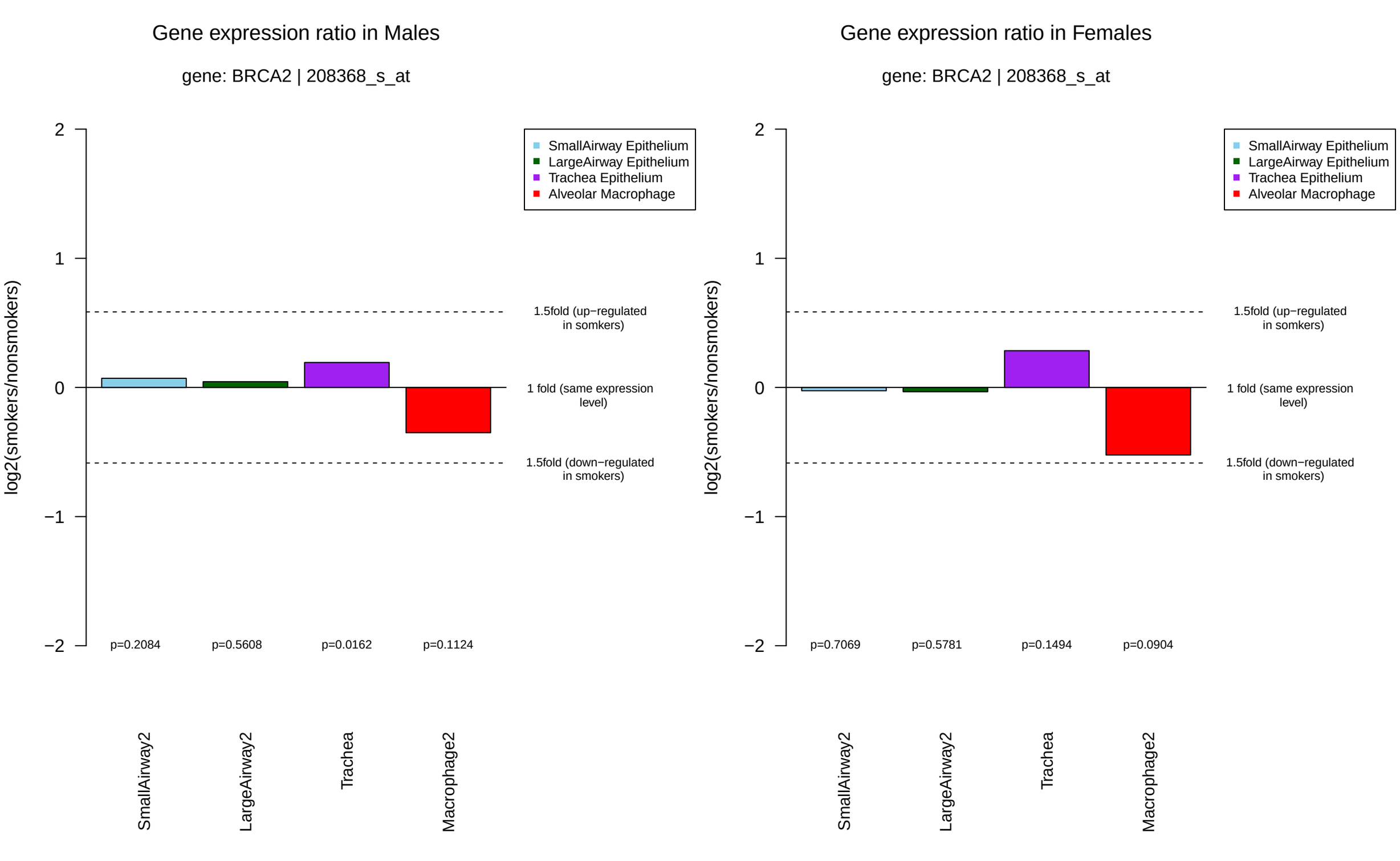
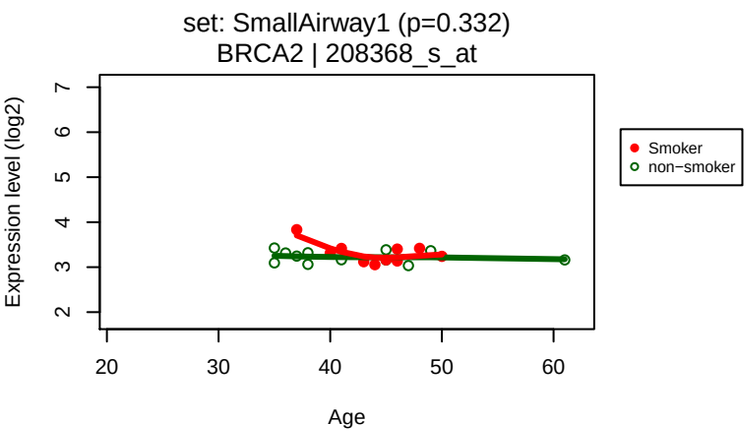
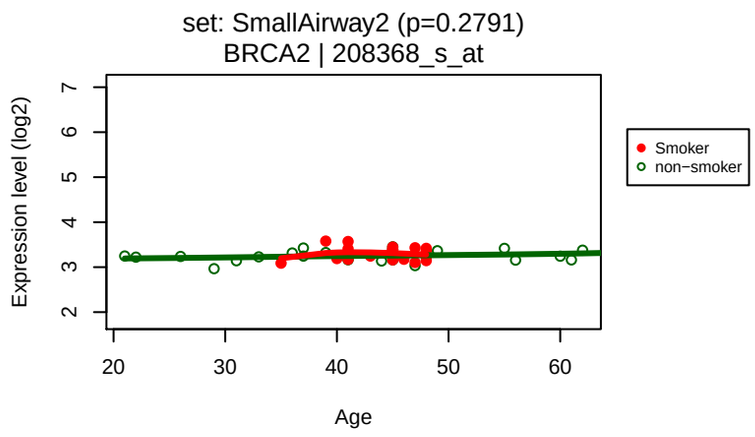
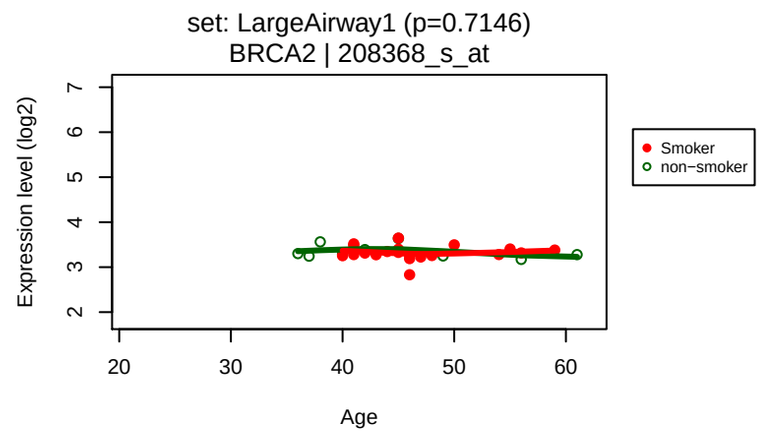

Gene expression of smokers and nonsmokers
gene: BRCA2 | 208368_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.332
p=0.2791
p=0.7849
p=0.7146
p=0.6576
p=0.0059
p=0.0303
p=0.0201
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

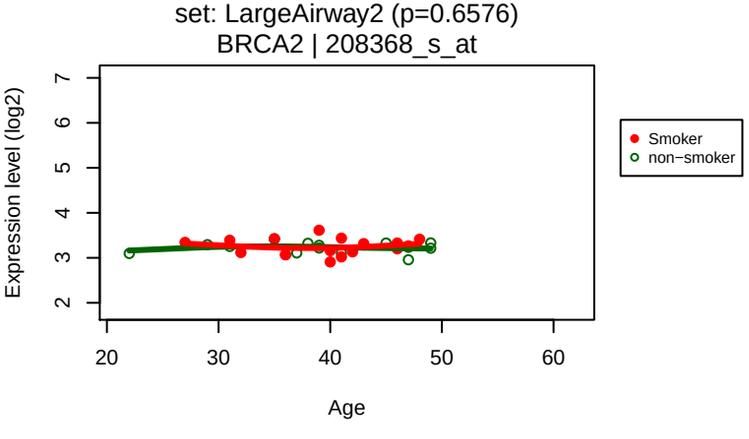

Gene expression ratio (smokers/nonsmokers)
gene: BRCA2 | 208368_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.332
p=0.2791
p=0.7849
p=0.7146
p=0.6576
p=0.0059
p=0.0303
p=0.0201
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

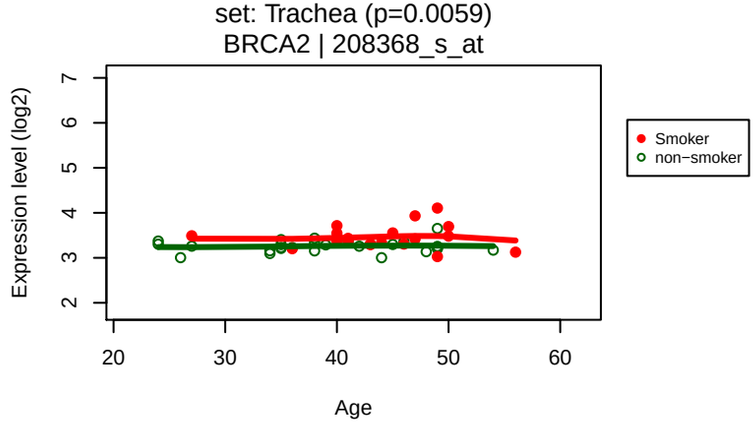

Gene expression ratio in Males
gene: BRCA2 | 208368_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2084
p=0.5608
p=0.0162
p=0.1124
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRCA2 | 208368_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7069
p=0.5781
p=0.1494
p=0.0904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

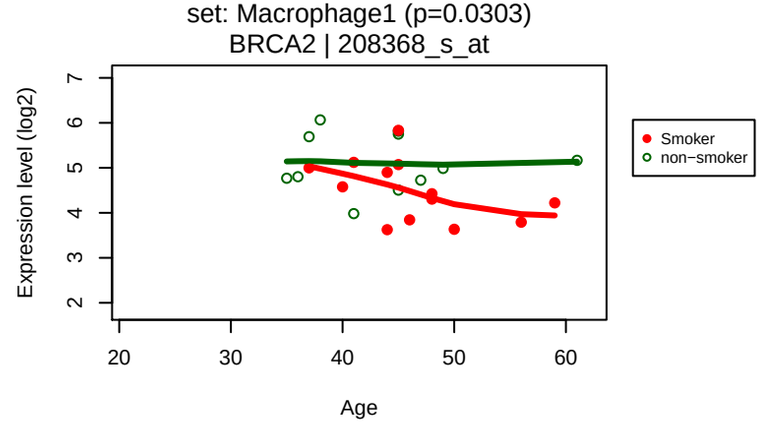

set: SmallAirway1 (p=0.332)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker

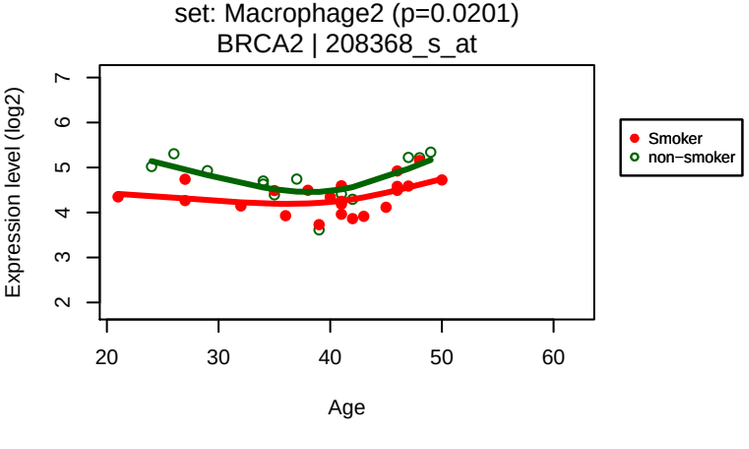

set: SmallAirway2 (p=0.2791)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker

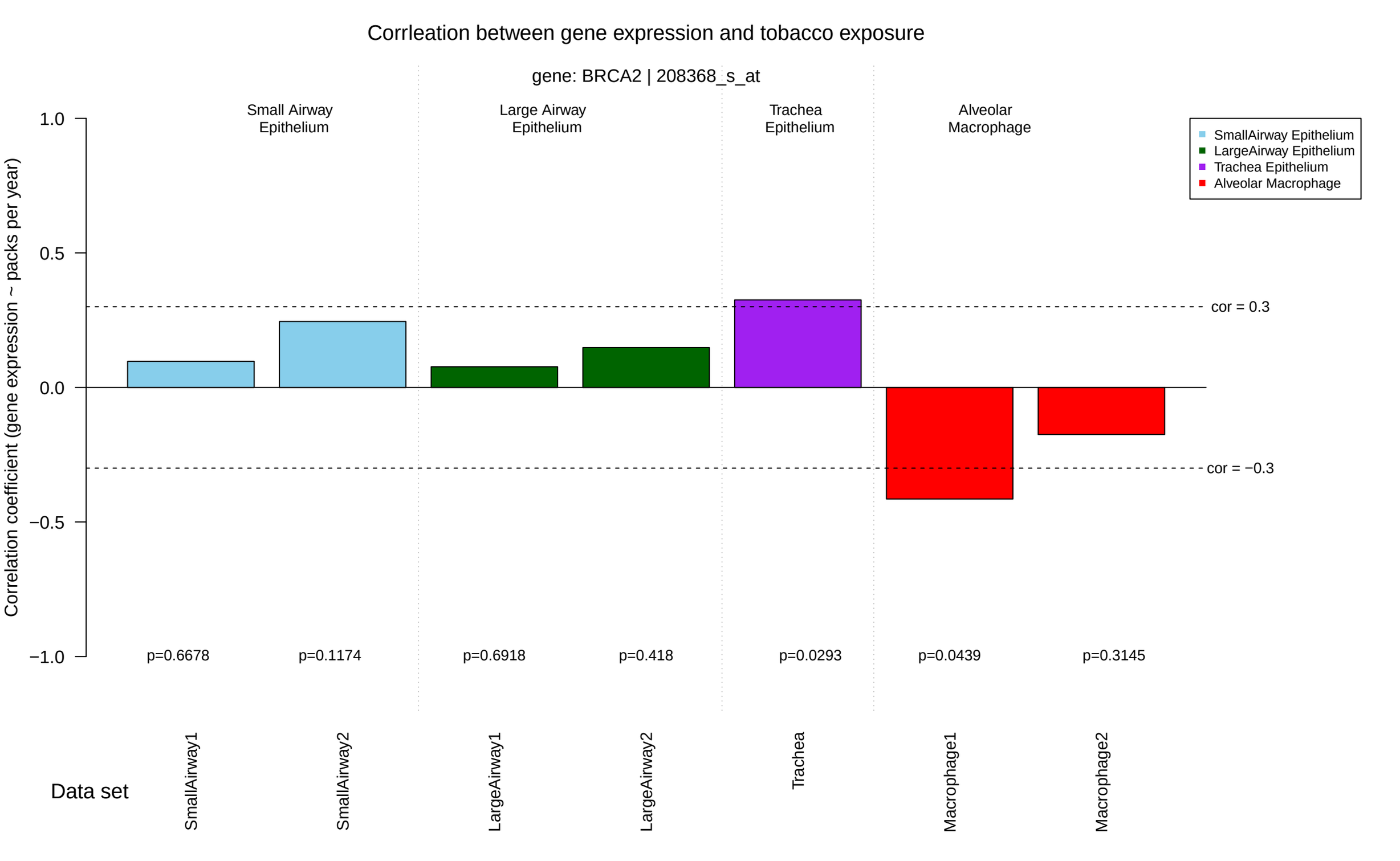

set: LargeAirway1 (p=0.7146)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6576)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0303)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0201)
BRCA2 | 208368_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRCA2 | 208368_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6678
p=0.1174
p=0.6918
p=0.418
p=0.0293
p=0.0439
p=0.3145
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2